Naïve Bayes (Continued)

↳ Using Naive Bayes, it is easier to classify many emails but what if a new word comes?

Ex- You won a Jackpot ⇝ If this was in the training data, it is easier to classify

Naive Bayes calculates the prob. of each word

Now, suppose email is changed to →

"You won a Lottery" ⇝ changed by spammers

↓

New for algorithm → may be classified as not spam

$$P(\text{You}|y=1) \cdot P(\text{won}|y=1) \cdot P(\text{a}|y=1) \cdot P(\text{Lottery}|y=1)$$

$$0.025 \times 0.05 \times 0.17 \times 0 = 0$$

↓

entire O/P is zero

So, to overcome this issue →

$$P(w|y) = \frac{n_1}{n}$$

where, n_1 = no. of times word 'w' occurs in the dataset

n = total no. of words

$$P(w|y) = \frac{n_1 + \alpha}{n + 2\alpha}$$

where, α = constant which is known as laplace smoothing

as Laplace smoothing
or
smoothing constant

∴ α acts as a hyperparameter

And we will use this smoothing constant with every word

★ Hyperparameter Tuning :-

① $\alpha = 10000$

$$P(w|y) = \frac{n_1 + \alpha}{n + 2\alpha}$$

Let, $n_1 = 10$
$n = 100$

$$= \frac{10 + 10000}{100 + 20000} = \frac{10010}{20100}$$

$$= 0.49 \approx 0.5 \text{ or } 50\%$$

50% prob → model is not learning anything
underfitting

② $\alpha = 0.1$

$$P(w|y) = \frac{n_1 + \alpha}{n + 2\alpha}$$

$$= \frac{10 + 0.1}{100 + 0.2}$$

$$= \frac{10.1}{100.2} = 0.1 \text{ or } 10\%$$

$\alpha \downarrow\downarrow \rightsquigarrow$ 10% prob.

$\alpha \downarrow\downarrow \rightsquigarrow$ 10% prob.

③ $\alpha = 0$

$$P(w|y) = \frac{n_1 + \alpha}{n + 2\alpha}$$

$$= \frac{10 + 0}{100 + 0} = \frac{10}{100} = 0.1 = 10\%$$

$\alpha = 0$ is same as very small α, giving low prob. which does not make any impact on the actual prob.

∴ to generalize we can call it overfitting in some cases.

So, $\alpha \downarrow\downarrow \rightarrow$ overfitting (some cases)

$\alpha \uparrow\uparrow \rightarrow$ underfitting

Till now, we were calculating the prob. of each word by looking at the presence or absence of a word.

But,

for finding out how many times a word appears in an email we use

"Multinomial Naive Bayes"

Ex-

	w_1	w_2	w_3	w_4	w_5	
E_1	2	0	1	3	0	
E_2	0	4	1	0	2	
E_3	0	2	5	3	0	

Till Now, $P(w_1 | y=1)$, $P(w_1 | y=0)$ } Bernoulli NB

But, for Multinomial NB →

$P(w_1 = 1 | y=1)$, $P(w_1 = 2 | y=1)$, $P(w_1 = 3 | y=1)$ ----

$P(w_1 = 1 | y=0)$, $P(w_1 = 2 | y=0)$, $P(w_1 = 3 | y=0)$ ----

Here, we will calculate multiple probabilities of each word depending upon the frequency conditioned on class.

∴ Final Equation :-

$$P(y=1 | t) = P(y=1) \cdot \sum_{i=1}^{n} \log [P(w_i = K | y=1)]$$

$$P(y=0 | t) = P(y=0) \cdot \sum_{i=1}^{n} \log [P(w_i = K | y=0)]$$

* Feature Importance :-

↳ Feature / Word ⇝ Probability ↑

↳ Important feature

↳ That particular word is appearing frequently

* Limitations of NB :-

1) Computationally Expensive

Limitations of NB:

1. Computationally Expensive
2. Slower
3. So many mathematical assumptions

* Gaussian NB:-

When there are numerical features rather than categorical features then in such cases we use gaussian NB.